

Learning Without Backpropagation: A Controlled Comparison of Feedback Alignment, Target Propagation, Forward-Forward, Equilibrium Propagation and Weight Perturbation

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September 29, 2026

Abstract

Backpropagation (BP) is the workhorse of deep learning, yet it is widely regarded as biologically implausible and awkward on neuromorphic hardware because it needs weight transport, a separate backward phase and global, non-local error signals. A large literature proposes alternatives. We review six of them—Feedback Alignment (FA), Direct Feedback Alignment (DFA), Difference Target Propagation (DTP), the Forward-Forward algorithm (FF), Equilibrium Propagation (EqProp) and weight perturbation (SPSA)—and compare them with BP in a controlled benchmark: the same 784-256-128-10 network, data split, optimizer family, learning-rate schedule and validation-based tuning protocol, with every update rule implemented by hand and without autograd. Beyond accuracy we measure convergence speed, generalisation gap, measured multiply-accumulate cost, robustness to depth and to small data, and—for each method—the cosine alignment between its update and the true gradient. On MNIST, BP reaches 98.5%; Equilibrium Propagation comes within 0.6 points but costs $28\times$ more multiply-accumulates, FA and DFA lose about one point, DTP and Forward-Forward lose 2.5–4 points, and weight perturbation with 16 directions reaches 90.8%. Measured gradient alignment explains part of this: hidden-layer updates of FA/DFA have cosine ≈ 0.4 with the true gradient, and the alignment of SPSA matches the theoretical $\sqrt{k/(d+k)}$ law. With depth, BP is unaffected while FA, DFA and DTP degrade markedly (6 hidden layers: 89%, 92%, 26%). **[TODO: Fashion-MNIST numbers: run on Colab and update abstract.]**

1 Introduction

Error backpropagation [17] computes exact gradients by running the network’s computation graph backwards. Three properties of that procedure are problematic outside a GPU: (i) the backward pass uses the transpose of the forward weights (the *weight transport* problem [5, 3]); (ii) neurons must alternate between a forward and a backward phase and store forward activity until the error arrives; (iii) the update of a synapse depends on a signal that is computed far away in the network, so it is not *local*. Reviews of these issues from the neuroscience side include [20, 13]; from the machine-learning side, [1, 22, 2].

This paper does two things. First, Section 3 gives a unified description of six alternatives, expressed as the update rule each one applies to the same network. Second, Sections 4–5 report a controlled experimental comparison whose axes follow the properties above: accuracy, learning speed, computational cost, quality of the gradient estimate, scaling with depth, and data efficiency.

Contributions. (1) A single code base (PyTorch, no autograd in any learning rule) in which all methods share data, architecture family, optimizer and tuning protocol. (2) A direct measurement of gradient alignment for every method that has a gradient to compare with, including EqProp against back-propagation-through-time of its own relaxation. (3) An empirical study of how alignment, accuracy and cost change with depth, data size and (for SPSA and EqProp) the estimator’s hyper-parameters.

2 Background and taxonomy

Let a network have layers $a_l = f(z_l)$, $z_l = W_l a_{l-1} + b_l$, and loss $\mathcal{L}$. BP sets $\delta_L = \nabla_{z_L} \mathcal{L}$ and $\delta_l = (W_{l+1}^\top \delta_{l+1}) \odot f'(z_l)$, $\Delta W_l = -\eta \delta_l a_{l-1}^\top$. Table 1 classifies the alternatives by which of BP’s requirements they remove.

Method	Weight transport	Separate backward pass	Update local in space	Estimator of $\nabla \mathcal{L}$
BP	yes	yes	no	exact
FA	no (fixed random B)	yes	partly	biased, learned alignment
DFA	no	no (broadcast of e)	partly	biased
DTP	no (learned inverses)	feedback network	yes, given targets	biased
FF	no	no	yes (layer-wise objective)	different objective
EqProp	symmetric weights	no (second relaxation)	yes (Hebbian)	$\rightarrow$ exact as $\beta \rightarrow 0$
SPSA	no	no	no (global scalar)	unbiased, high variance

Table 1: Qualitative taxonomy of the compared learning rules.

3 Methods

All methods below train the MLP $784 \rightarrow 256 \rightarrow 128 \rightarrow 10$ (except where stated) and produce, for a mini-batch, an update direction that is passed to a common optimizer.

Feedback Alignment [12]. Replace $W_{l+1}^\top$ in the backward recursion by a fixed random matrix B_{l+1} : $\delta_l = (B_{l+1} \delta_{l+1}) \odot f'(z_l)$. The forward weights come to align with B so that the resulting update correlates positively with the true gradient [12, 16].

Direct Feedback Alignment [15]. The output error e is projected directly to every hidden layer, $\delta_l = (B_l e) \odot f'(z_l)$, removing the layer-to-layer backward chain [9].

Difference Target Propagation [11]. Hidden layers are trained by local regression $\frac{1}{2} \|h_l - \hat{h}_l\|^2$ to targets propagated downwards through learned approximate inverses g_l : $\hat{h}_{L-1} = h_{L-1} - \alpha \partial \mathcal{L} / \partial h_{L-1}$ and $\hat{h}_{l-1} = h_{l-1} + g_l(\hat{h}_l) - g_l(h_l)$. The inverses are trained with a noisy auto-encoding loss $\|g_l(f_l(h + \varepsilon)) - (h + \varepsilon)\|^2$. Hidden units are tanh. See [14] for a theoretical framing.

Forward-Forward [6]. Each layer optimises a local objective that separates *positive* data (image with its correct label written in the first ten pixels) from *negative* data (wrong label). With goodness $G_l = \frac{1}{n_l} \sum_j h_{l,j}^2$ and threshold θ , the layer minimises $\log(1 + e^{-(G_l^+ - \theta)}) + \log(1 + e^{G_l^- - \theta})$; layer inputs are length-normalised so no information passes via goodness. Prediction: the label whose

embedding maximises goodness summed over layers $2..L$. FF has no output layer, so it uses 3×500 hidden units.

Equilibrium Propagation [18]. The network is an energy-based model with symmetric couplings and hard-sigmoid units $\rho(s) = \text{clip}(s, 0, 1)$. The free phase relaxes to an equilibrium s^0 ; in the nudged phase the output is weakly pushed toward the target by $\beta(y - s_L)$, giving s^β . The update $\Delta W_l \propto \frac{1}{\beta}(\rho(s_{l-1}^\beta)^\top \rho(s_l^\beta) - \rho(s_{l-1}^0)^\top \rho(s_l^0))$ estimates $-\partial C / \partial W_l$ and becomes exact as $\beta \rightarrow 0$. We use the symmetric ($\pm\beta$) variant [8] and, as reference gradient, BPTT through the relaxation [4].

Weight perturbation / SPSA [19, 2]. $\hat{g} = \frac{1}{k} \sum_{i=1}^k \frac{\mathcal{L}(\theta + \epsilon v_i) - \mathcal{L}(\theta - \epsilon v_i)}{2\epsilon} v_i$, $v_i \sim \mathcal{N}(0, I_d)$. It is unbiased for $\nabla \mathcal{L}$ but its cosine to the true gradient scales as $\sqrt{k/(d+k)}$ for isotropic gradients, so it needs many directions in high dimension.

4 Experimental setup

Data. MNIST [10] and Fashion-MNIST [21]; 50 000/10 000/10 000 train/validation/test; pixels in $[0, 1]$.

Protocol. Batch 128, 20 epochs, cosine learning-rate decay to zero for every method, Adam [7] or SGD with momentum as selected on the validation set. Per-method hyper-parameters (learning rate and method-specific knobs) are found by random search (10 trials, 5 epochs, half of the training set) that never touches the test set; then each method is run with seeds 0,1,2 and we report mean $\pm$ std of the final-epoch test accuracy.

Cost. All matrix products go through one counter, so “MACs per sample” is measured, not estimated; wall-clock is reported for a fixed 2-vCPU CPU machine (PyTorch, float32); **[TODO: re-measure on a GPU]**.

Gradient alignment. At the end of every epoch, on a fixed batch of 1000 training images, we compute the cosine between the method’s update and the reference gradient (exact BP for the MLP methods; BPTT through the relaxation for EqProp). FF is excluded since it optimises a different objective.

5 Results

5.1 Accuracy, speed and generalisation

BP reaches $98.47 \pm 0.06\%$ test accuracy. EqProp is the closest alternative ($97.91 \pm 0.14\%$), followed by FA (97.53%) and DFA (97.50%), DTP (95.96%), FF (94.35%) and SPSA (90.76%). All methods except FF and SPSA fit the training set almost perfectly ($\geq 97.5\%$ train accuracy), so the differences are mainly differences in how well the update direction optimises. The train–test gap is 1.5 points for BP, 1.6 for DTP, 1.9 for EqProp, 2.0 for FA and 2.2 for DFA, i.e. the alternatives with more aligned-but-biased updates do not generalise better than BP here. FF (which uses 3×500 hidden units, i.e. $3.8\times$ the parameters) under-fits (train 94.7%, gap 0.36 points), consistent with public re-implementations, and is below the numbers of the original paper, which uses larger networks and more training. SPSA is also under-fitted and, importantly, was the only method whose stability

Dataset	Method	Test acc. (%)	Train–test gap	Epochs to 90%	MACs/sample (M)	s/epoch	cos to grad.
MNIST	Backprop	98.47 ± 0.06	1.53	1.0	0.50	1.6	1.00
MNIST	Feedback Alignment	97.53 ± 0.02	2.04	1.3	0.50	1.6	0.96
MNIST	Direct FA	97.50 ± 0.08	2.21	2.3	0.47	1.6	0.84
MNIST	Diff. Target Prop	95.96 ± 0.07	1.57	1.0	0.63	1.7	0.66
MNIST	Forward-Forward	94.35 ± 0.16	0.36	1.3	3.57	6.8	nan
MNIST	Equilibrium Prop	97.91 ± 0.14	1.87	1.0	14.14	12.8	0.97
MNIST	SPSA (weight pert.)	90.76 ± 0.12	-0.61	8.3	7.51	14.8	0.01

Table 2: Main benchmark (mean over 3 seeds). “Epochs to 90%” is the first epoch with validation accuracy $\geq 90\%$.

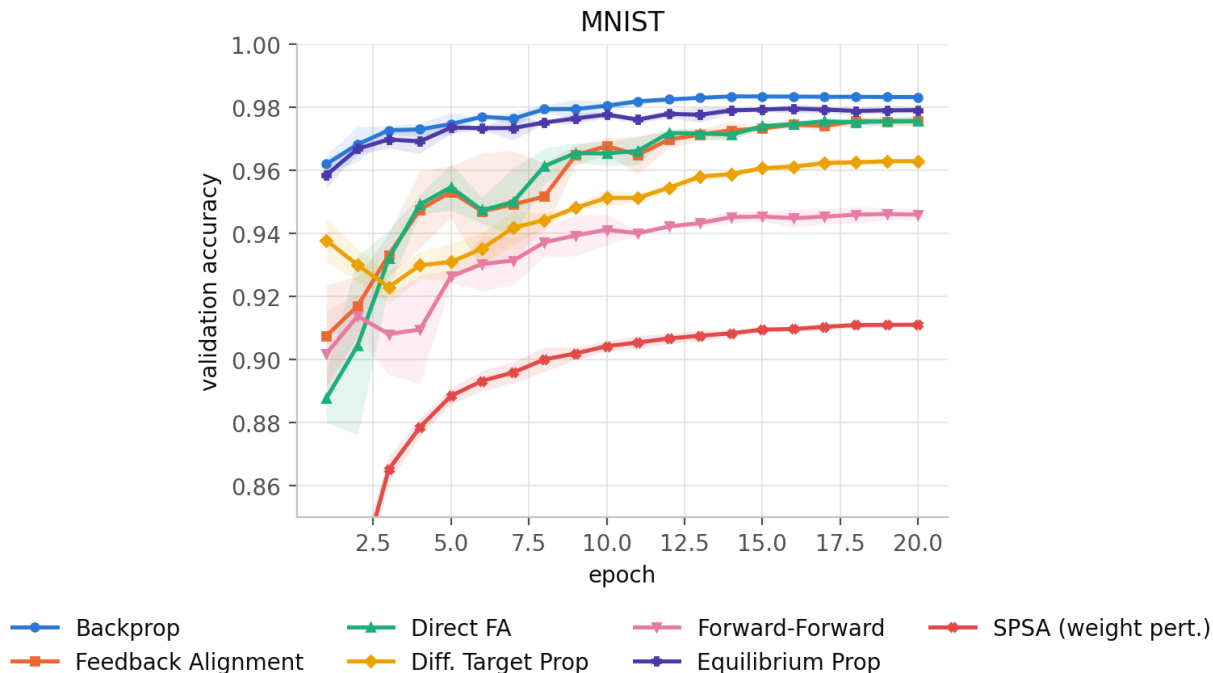


Figure 1: Validation accuracy per epoch (mean $\pm$ std over seeds).

required an extra device (update-norm clipping): with $k = 16$ its update has, in expectation, a norm $\approx \sqrt{d/k} \approx 120$ times that of the true gradient. [TODO: Fashion-MNIST: repeat this paragraph.]

5.2 Quality of the update direction

Starting from an alignment of 0.14 at initialisation, FA reaches 0.95 within the first epoch and stays at 0.96, in line with the “align, then memorise” picture [16]. The global number is misleading, however: the output-layer update of FA and DFA is exact, and it dominates the gradient norm. Per layer (Fig. 3) the hidden-layer cosines at the end of training are only 0.37/0.36 (FA) and 0.50/0.37 (DFA). DTP’s top hidden layer receives a true gradient step by construction (cosine 1.0) while its first hidden layer aligns only at 0.26. EqProp’s update is the only one aligned across all layers (0.97–0.98), but this alignment fluctuates during training between 0.5 and 0.97: with the tuned $\beta = 0.25$ the estimator is biased (Fig. 4, left: cosine 0.63 at $\beta = 0.25$, 0.91 at 0.1, 0.998 at 0.01), yet the method still trains well, so *exactness of the gradient is not necessary for good accuracy*. SPSA’s alignment is 0.006 (0.05 for the 1290-parameter output layer), matching $\sqrt{k/(d+k)}$ (Fig. 4, right).

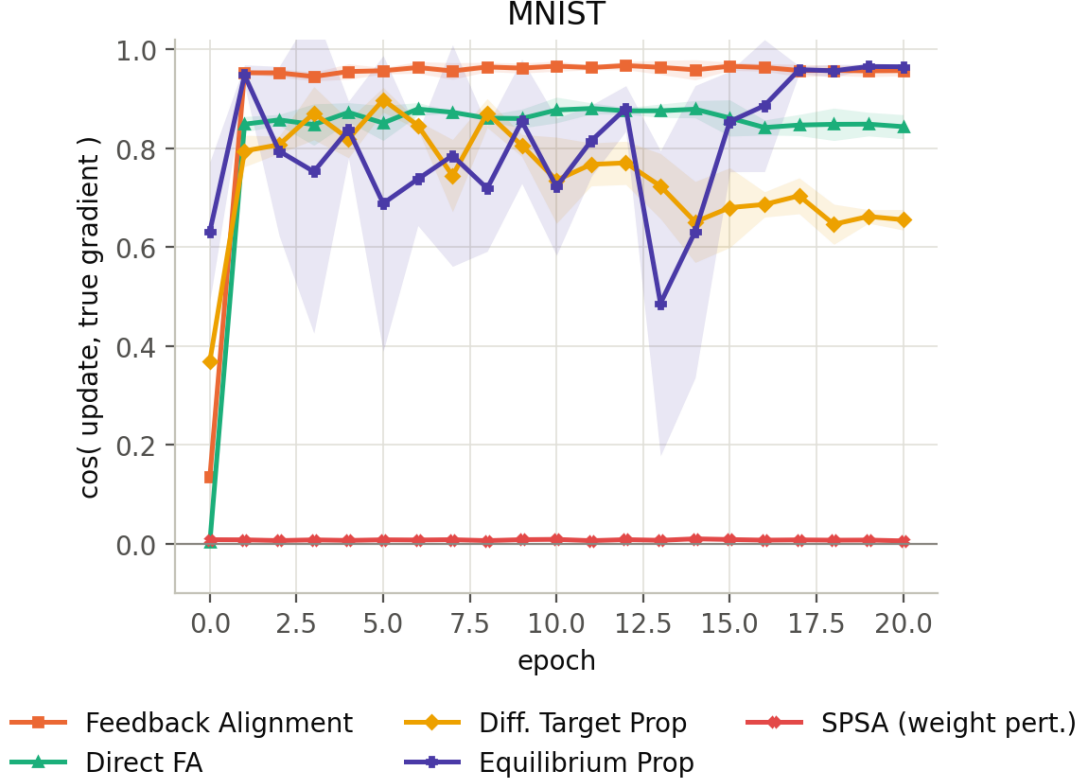


Figure 2: Cosine between each method’s update and the true gradient during training.

5.3 EqProp converges to the gradient; SPSA needs many directions

5.4 Depth and data efficiency

With 2/3/4/6 hidden layers BP stays at 98.3–98.4% (2–4 layers) and 98.2% (6). FA drops 97.1 $\rightarrow$ 95.2 $\rightarrow$ 93.7 $\rightarrow$ 89.2%, DFA 97.2 $\rightarrow$ 95.7 $\rightarrow$ 94.1 $\rightarrow$ 92.3%, DTP 95.4 $\rightarrow$ 91.3 $\rightarrow$ 82.3 $\rightarrow$ 26.0%. EqProp is competitive up to 4 layers (98.0, 97.7, 97.3%) but with fixed hyper-parameters (relaxation length scaled with depth) one of two seeds fails to train at 6 layers (mean 58.8%), we did not diagnose the cause (a natural suspect is loss of stability of the relaxation dynamics in deeper coupled networks; checking the spectral radius of the Jacobian is left to future work). Data efficiency (fixed number of gradient steps): with 1000 examples the spread between best and worst method is 4.4 points (89.7% BP vs. 85.3% FF); with 50 000 it is 9.7 points; SPSA barely improves with more data (85.9 $\rightarrow$ 88.7%), since it is optimisation-limited.

5.5 Computational cost

6 Discussion

Provably vs. empirically convergent. EqProp is the only method with a convergence guarantee to the true gradient (as $\beta \rightarrow 0$), and we confirm it numerically. FA has none, but empirically aligns fast in shallow nets and degrades with depth. In practice both regimes coexist: the “provable” method was used at a large β where it is biased, and still worked best of the alternatives.

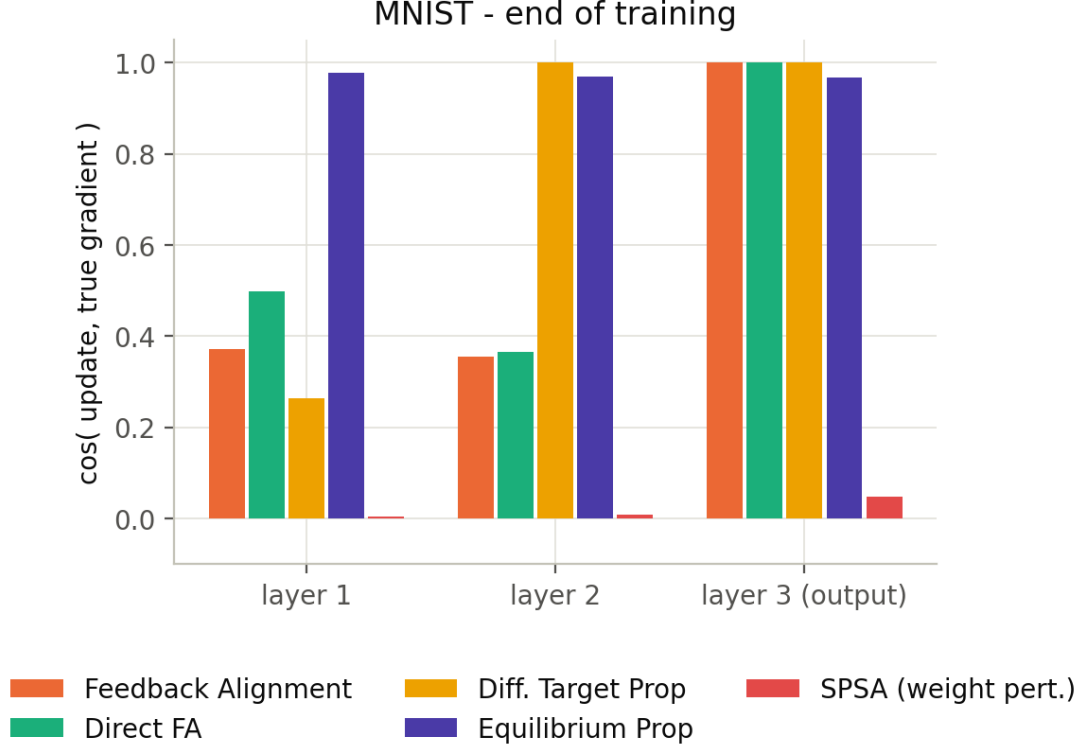


Figure 3: Per-layer alignment at the end of training (layer 3 is the output layer, whose update is exact for FA/DFA).

Cost. Locality is not free. EqProp needs a relaxation (here $30 + 2 \times 10$ steps) per update, $28 \times$ the MACs of BP; SPSA needs $2k$ forward passes ($15 \times$); FF processes positive and negative data and 10 label hypotheses at inference. FA/DFA cost the same as BP on a GPU; their advantage is hardware-side (no weight transport), not FLOPs.

Limitations. A single small MLP, two datasets, three seeds (two for depth/data), one hyper-parameter budget; no convolutional or large-scale experiments, where FA/DTP/FF are known to be much weaker [1, 22, 9]. Hyper-parameters were tuned on MNIST and (unless re-tuned) reused for Fashion-MNIST. [TODO: Hebbian/STDP learning is not covered.]

7 Conclusion

In a controlled small-scale comparison, no alternative matches BP at equal cost; the closest (EqProp) pays with an order of magnitude more computation, and the cheap ones (FA, DFA) lose accuracy that grows with depth because their hidden-layer updates are only weakly aligned with the gradient. Methods that dispense with a global error path entirely (FF, SPSA) trail by 4–8 points on MNIST. [TODO: Fashion-MNIST sentence.]

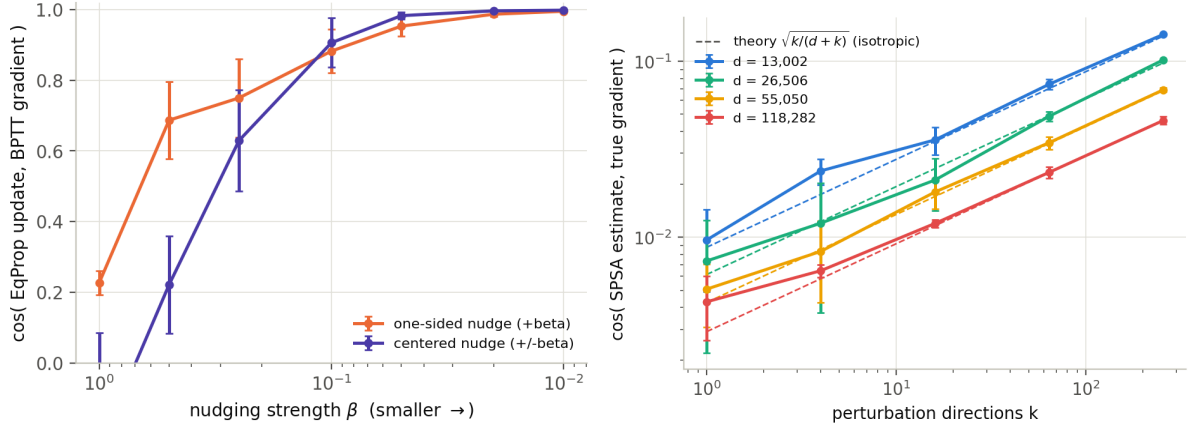


Figure 4: Left: EqProp update vs. BPTT gradient as $\beta \rightarrow 0$ (theory: exact in the limit). Right: SPSA alignment vs. number of directions k for several parameter counts d ; dashed: $\sqrt{k/(d+k)}$.

References

- [1] S. Bartunov, A. Santoro, B. A. Richards, L. Marris, G. E. Hinton, and T. Lillicrap. Assessing the scalability of biologically-motivated deep learning algorithms and architectures. In *NeurIPS*, 2018.
- [2] A. G. Baydin, B. A. Pearlmutter, D. Syme, F. Wood, and P. Torr. Gradients without back-propagation, 2022.
- [3] F. Crick. The recent excitement about neural networks. *Nature*, 337:129–132, 1989.
- [4] M. Ernoult, J. Grollier, D. Querlioz, Y. Bengio, and B. Scellier. Updates of equilibrium prop match gradients of backprop through time in an RNN with static input. In *NeurIPS*, 2019.
- [5] S. Grossberg. Competitive learning: from interactive activation to adaptive resonance. *Cognitive Science*, 11:23–63, 1987.
- [6] G. Hinton. The forward-forward algorithm: some preliminary investigations, 2022.
- [7] D. P. Kingma and J. Ba. Adam: a method for stochastic optimization. In *ICLR*, 2015.
- [8] A. Laborieux, M. Ernoult, B. Scellier, Y. Bengio, J. Grollier, and D. Querlioz. Scaling equilibrium propagation to deep ConvNets by drastically reducing its gradient estimator bias. *Frontiers in Neuroscience*, 15, 2021.
- [9] J. Launay, I. Poli, F. Boniface, and F. Krzakala. Direct feedback alignment scales to modern deep learning tasks and architectures. In *NeurIPS*, 2020.
- [10] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.
- [11] D.-H. Lee, S. Zhang, A. Fischer, and Y. Bengio. Difference target propagation. In *ECML PKDD*, 2015.

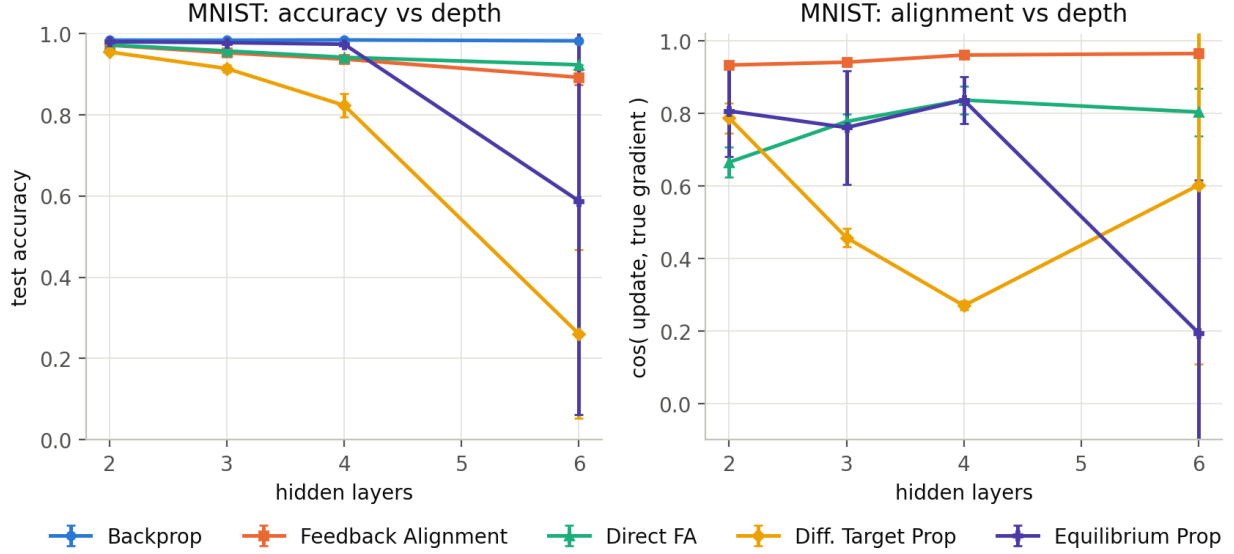


Figure 5: Accuracy and alignment vs. number of hidden layers.

- [12] T. P. Lillicrap, D. Cownden, D. B. Tweed, and C. J. Akerman. Random synaptic feedback weights support error backpropagation for deep learning. *Nature Communications*, 7:13276, 2016.
- [13] T. P. Lillicrap, A. Santoro, L. Marris, C. J. Akerman, and G. Hinton. Backpropagation and the brain. *Nature Reviews Neuroscience*, 21:335–346, 2020.
- [14] A. Meulemans, F. S. Carzaniga, J. A. K. Suykens, J. Sacramento, and B. F. Grewe. A theoretical framework for target propagation. In *NeurIPS*, 2020.
- [15] A. Nøkland. Direct feedback alignment provides learning in deep neural networks. In *NeurIPS*, 2016.
- [16] M. Refinetti, S. d’Ascoli, R. Ohana, and S. Goldt. Align, then memorise: the dynamics of learning with feedback alignment. In *ICML*, 2021.
- [17] D. E. Rumelhart, G. E. Hinton, and R. J. Williams. Learning representations by back-propagating errors. *Nature*, 323:533–536, 1986.
- [18] B. Scellier and Y. Bengio. Equilibrium propagation: bridging the gap between energy-based models and backpropagation. *Frontiers in Computational Neuroscience*, 11:24, 2017.
- [19] J. C. Spall. Multivariate stochastic approximation using a simultaneous perturbation gradient approximation. *IEEE Transactions on Automatic Control*, 37(3):332–341, 1992.
- [20] J. C. R. Whittington and R. Bogacz. Theories of error back-propagation in the brain. *Trends in Cognitive Sciences*, 23(3):235–250, 2019.
- [21] H. Xiao, K. Rasul, and R. Vollgraf. Fashion-MNIST: a novel image dataset for benchmarking machine learning algorithms, 2017.
- [22] W. Xiao, H. Chen, Q. Liao, and T. Poggio. Biologically-plausible learning algorithms can scale to large datasets, 2018.

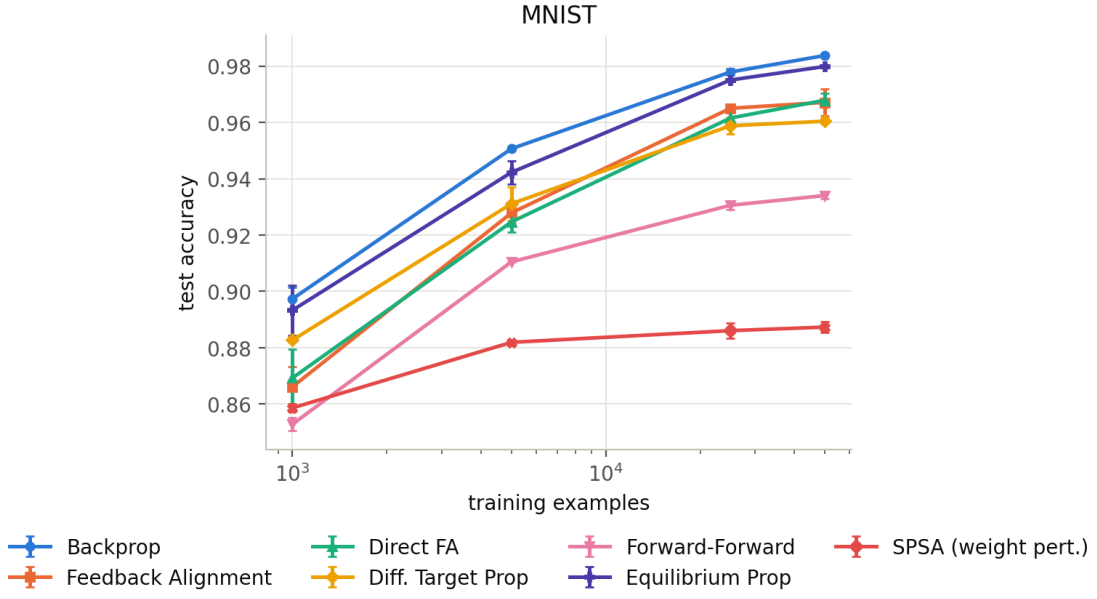


Figure 6: Test accuracy vs. training-set size at a fixed number of gradient steps.

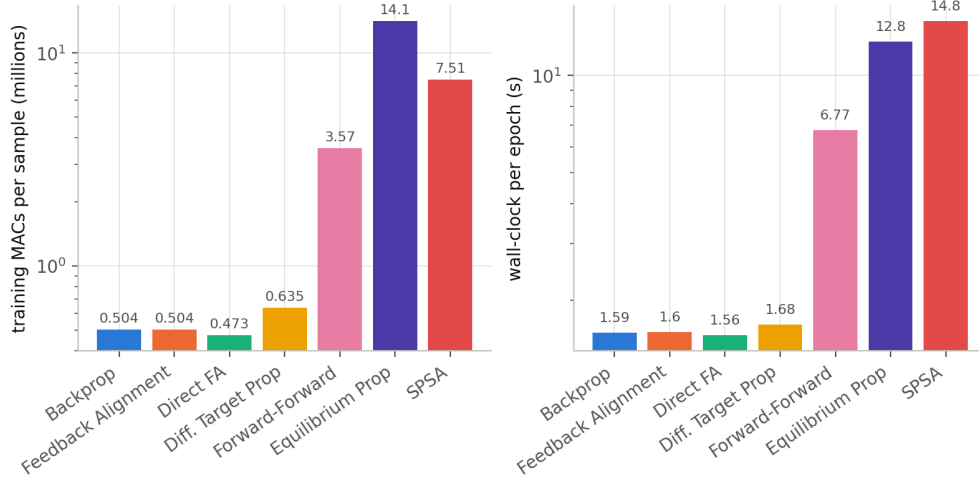


Figure 7: Measured training cost per sample.

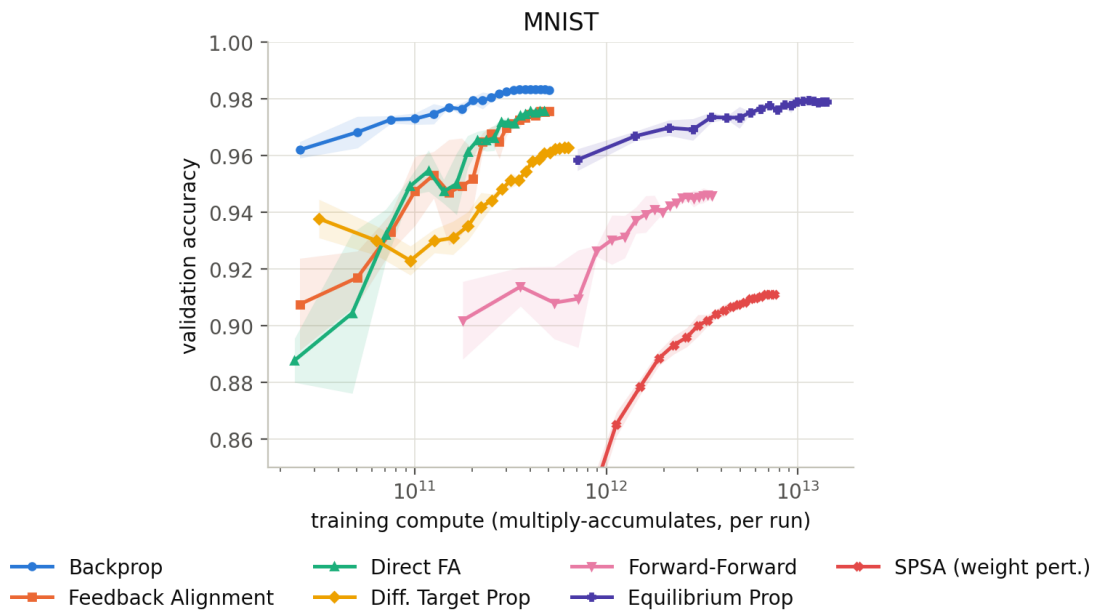


Figure 8: Validation accuracy against cumulative multiply-accumulates.